

DEBIASITYOURSELF: Structured Supervision for Generalizable Bias-Reducing Interventions

Anonymous ACL submission

Abstract

Large language models often handle bias-sensitive context as if social salience were task evidence: names, group descriptors, roles, occupations, events, and institutions can become shortcuts even when they are irrelevant to the correct answer. We introduce DEBIASITYOURSELF, a structured supervision method for teaching reusable bias-reducing behavior. DEBIASITYOURSELF translates prejudice-reducing interventions from social psychology into model-facing supervision, covering STEREOTYPE REPLACEMENT, COUNTER-STEREOTYPIC IMAGING, INDIVIDUATION, PERSPECTIVE-TAKING, and POSITIVE CONTACT. The contribution is not a new prompt template or a psychology label attached to fine-tuning; the intervention inventory defines a shared training and inference schema for identifying unsupported social assumptions and producing task-consistent outputs. We evaluate DEBIASITYOURSELF across CrowS-Pairs, StereoSet, BBQ, WinoBias, and WinoGender, using normalized bias error so that lower values consistently indicate better mitigation. We compare DEBIASITYOURSELF IT, ICL, and DEBIASITYOURSELF two-pass inference against a broad set of debiasing baselines, and we evaluate Balanced COPA and ARC to test whether bias reduction preserves reasoning competence. Across the current results, DEBIASITYOURSELF improves cross-benchmark bias error most consistently when intervention-guided inference is paired with structured supervision, while companion QWEN and LLAMA-70B analyses suggest that the same pattern is not tied to a single model family.

1 Introduction

Language models are trained on text that is saturated with social information. Names, demographic descriptors, occupations, institutions, events, neighborhoods, dialectal cues, and interpersonal roles all appear in contexts shaped by historical asymmetries and cultural stereotypes. These cues are

often useful for understanding a task, but they can also become shortcuts. A model may treat a role as evidence of competence, an occupation as evidence of gender, a group descriptor as evidence of behavior, or a social situation as evidence for blame. The resulting failure is not only that models encode biased associations. A model may know that stereotyping is undesirable and still fail to decide, in context, whether a social cue is relevant evidence, irrelevant background, or an unsupported heuristic.

This distinction matters for bias mitigation. A good model should not simply erase social information or refuse to answer whenever a sensitive cue appears. Sometimes the cue is irrelevant and should not affect the answer. Sometimes it is relevant but must be grounded in the evidence provided by the task. Sometimes the input invites a stereotyped inference that should be revised. In other cases, the model should preserve the task objective while avoiding generic moralizing or over-sanitized language. Across likelihood benchmarks, ambiguous question answering, coreference, and generation, the same underlying challenge appears: the model must handle bias-sensitive context without letting social salience become social evidence.

Most existing mitigation pipelines intervene through a particular mechanism: counterfactual augmentation, parameter-efficient fine-tuning, activation steering, decoding constraints, preference optimization, or prompt-based self-revision. These mechanisms are valuable, but the mechanism itself is rarely a sufficient theory of the behavior the model should learn. A method that improves a pairwise likelihood benchmark may not teach the model how to answer an ambiguous question. A method that reduces a stereotype score may do so by becoming vague, evasive, or less useful. A method tuned around one bias category may not define what to do when the cue changes from an identity term to an occupation, a social role, or a

contextual relation.

DEBIASITYOURSELF starts from a different technical premise: the core object of debiasing should be the bias-reducing behavior the model learns, not the delivery mechanism used to induce it. We define bias mitigation as learning intervention-guided response generation. Given an input that may invite an unsupported social shortcut, the model must produce an output justified by task evidence rather than by stereotype, salience, or majority priors. The intervention inventory is motivated by the prejudice habit-breaking intervention of Devine et al. (2012), which teaches concrete bias-reducing procedures such as STEREOTYPE REPLACEMENT, COUNTER-STEREOTYPIC IMAG-ING, INDIVIDUATION, PERSPECTIVE-TAKING, and POSITIVE CONTACT. We do not claim that language models possess implicit attitudes in the human psychological sense. Instead, we translate these procedures into model-facing supervision: each intervention specifies what evidence to attend to, what heuristic to avoid, and what kind of task-consistent output to produce.

DEBIASITYOURSELF implements this premise as a single structured supervision method. It constructs a broad set of biased and debiased instances, labels each debiased target with the intervention that explains the desired behavior, and uses the resulting data to teach intervention-guided revision and response generation. Prompting, fine-tuning, soft prompting, and self-revision are treated as interfaces to the same supervision signal. This distinction matters for the paper’s claim: the contribution is not psychology plus fine-tuning, but a method for teaching models how to handle bias-sensitive context through reusable bias-reducing interventions.

Our central claim is deliberately scoped:

Paper thesis. DEBIASITYOURSELF teaches language models to handle bias-sensitive context as evidence-grounded input rather than as a social shortcut, using reusable intervention-guided behaviors.

The paper evaluates this claim through three questions:

1. Does DEBIASITYOURSELF reduce bias across multiple datasets rather than only on the data format used to construct its supervision?
2. Which components matter: intervention labels, mixtures of interventions, rationales, and the interface used to provide supervision?
3. Does DEBIASITYOURSELF preserve reasoning

performance, or does stronger debiasing trade away task competence?

We make four contributions.

1. We formalize bias mitigation as structured debiasing supervision, rather than as a benchmark-specific post-hoc patch.
2. We construct a data generation pipeline that pairs biased instances with intervention-labeled debiased targets across social categories, instance types, and domains.
3. We instantiate DEBIASITYOURSELF through training-time and inference-time interfaces, using these variants as ablations of one method rather than separate methods.
4. We compare DEBIASITYOURSELF against a broad set of debiasing baselines across bias and reasoning benchmarks, explicitly measuring generalization across evaluation settings and utility preservation.

2 Related Work

Bias benchmarks. CrowS-Pairs evaluates whether a model assigns higher likelihood to stereotypical than anti-stereotypical sentence pairs across multiple social bias categories (Nangia et al., 2020). StereoSet measures stereotypical associations while also tracking language modeling ability through ICAT-style scoring (Nadeem et al., 2021). BBQ evaluates social bias in question answering by comparing model behavior in ambiguous and disambiguated contexts (Parrish et al., 2022). WinoBias and WinoGender study gender bias in coreference and occupation-linked schemas (Zhao et al., 2018; Rudinger et al., 2018). Additional evaluations such as HONEST, UNQOVER, BOLD, and Bias in Bios broaden the task surface to hurtful completions, underspecified questions, open-ended generation, and occupation prediction (Nozza et al., 2021; Li et al., 2020; Dhamala et al., 2021; De-Arteaga et al., 2019). DEBIASITYOURSELF uses this diversity as an evaluation requirement: a mitigation method should not be judged from one benchmark alone.

Bias mitigation in language models. Prior work has explored mitigation at several points in the modeling pipeline, including data augmentation, representation editing, parameter-efficient fine-tuning, activation steering, preference optimization, prompt-based revision, and decoding-time control.

Self-diagnosis and self-debiasing methods ask models to recognize and reduce their own toxic or biased continuations without additional training data (Schick et al., 2021). Empirical surveys show that debiasing methods can produce inconsistent effects across benchmarks and may reduce language modeling or downstream performance (Meade et al., 2021). DEBIASITYOURSELF is complementary to the mechanism-level view: it specifies the supervision signal a mitigation method should induce. The implementation mechanism is then an interface to the same structured supervision, not the main scientific unit.

Identity coverage. Many bias datasets focus on a small number of axes, often gender, race, or religion. HolisticBias expands descriptor coverage to a broader set of demographic axes and emphasizes participatory construction of identity descriptors (Smith et al., 2022). DEBIASITYOURSELF uses broad identity coverage during generation so that the intervention is not tied to one social category. This broad coverage is necessary for our generalization claim but also increases the need for careful safety review, because synthetic biased examples can reproduce harmful language even when used for mitigation.

Prejudice-reducing interventions as supervision design. The prejudice habit-breaking intervention treats implicit bias as a habit that can be changed through awareness, concern, and repeated use of bias-reducing procedures (Devine et al., 2012). The five interventions are not merely labels; they describe distinct behaviors for reducing biased responses. STEREOTYPE REPLACEMENT identifies and substitutes a biased response. COUNTER-STEREOTYPIC IMAGING makes non-stereotypical exemplars salient. INDIVIDUATION shifts attention from group membership to person-specific evidence. PERSPECTIVE-TAKING asks the reasoner to consider the affected person’s experience. POSITIVE CONTACT increases exposure to counter-stereotyped, constructive intergroup examples. DEBIASITYOURSELF adapts these intervention ideas into model-facing supervision, prompts, and inference-time revisions.

3 Method

3.1 Problem Formulation

Let x denote an input that may invite a stereotyped or unsupported group-based inference, and let y

denote the desired task output. Standard mitigation methods typically optimize for a narrow proxy: reduce a benchmark score, avoid a set of words, or steer a representation away from a bias direction. DEBIASITYOURSELF instead learns intervention-guided response generation:

$$p_{\theta}(y \mid x, b),$$

where b is a bias-reducing intervention label. The label specifies the desired debiasing behavior for the instance. In this formulation, a debiased output is not merely less toxic or less stereotypical; it must be consistent with the task evidence and the selected intervention.

The training signal for DEBIASITYOURSELF consists of triples (x, b, y_b) , where y_b is a debiased answer or revision produced under intervention b . Optional rationales r_b expose the intermediate reasoning:

$$x \xrightarrow{b} (r_b, y_b), \quad b \in \{\text{SR, CI, IND, PT, PC}\}.$$

The rationale is used as supervision or prompting context when helpful, but the central object is the conditional distribution over debiased outputs.

3.2 Bias-Reducing Interventions

The five interventions form a structured space of debiasing behaviors. They are deliberately overlapping: many biased instances can be revised in more than one way, but each intervention emphasizes a different failure mode. This gives DEBIASITYOURSELF two technical advantages. First, it can supervise multiple valid debiased outputs for the same social bias pattern. Second, it can expose which intervention transfers to which benchmark family.

3.3 Data Generation Pipeline

DEBIASITYOURSELF constructs intervention-labeled supervision in four stages. First, we build an identity inventory spanning multiple demographic dimensions, drawing from HolisticBias descriptors and generated identity lists. Second, we generate biased seed instances that cover several textual forms: direct statements, short stories, dialogue, news-style snippets, and questions. Third, for each seed instance we generate debiased targets under the five interventions. Fourth, we filter and format these triples into a unified instruction schema:

Input: x , Intervention: b ,
Debiased output: y_b .

Intervention	Human intervention idea	Model-facing behavior	Expected failure mode
STEREOTYPE REPLACEMENT	Notice a stereotypic response and replace it with a non-stereotypic response.	Identify biased association, label why it is unsupported, and rewrite or answer without the stereotype.	Overly generic refusal or moralizing without task completion.
COUNTER-STEREOTYPIC IMAGING	Make counter-stereotypic exemplars salient.	Introduce examples that violate the biased association while preserving task content.	Token replacement without deeper reasoning.
INDIVIDUATION	Attend to person-specific evidence rather than group-level assumptions.	Prefer contextual evidence over identity priors in QA, coreference, and likelihood comparisons.	Ignoring relevant identity context when it is genuinely task-relevant.
PERSPECTIVE-TAKING	Consider the experience of the affected person or group.	Simulate the impact of a biased assumption and revise the response accordingly.	Producing verbose empathy language instead of concise answers.
POSITIVE CONTACT	Increase constructive exposure to out-group members.	Generate balanced, non-tokenizing interactions involving identities across contexts.	Sanitized examples that lack stereotype contrast.

Table 1: The five bias-reducing interventions in DEBIASITYOURSELF. They define the structured supervision used for data generation, model conditioning, and ablation analysis. This table is a draft placeholder; final wording should be tightened after method review.

When rationales are used, they are represented as an auxiliary field rather than a separate method.

The generation pipeline explicitly separates three dimensions:

1. **Identity dimension:** age, disability, gender identity, nationality, physical appearance, race/ethnicity, religion, sexual orientation, socioeconomic status, and intersections available in the benchmark.
2. **Instance type:** opinion, action, event, story, dialogue, news-style text, or question.
3. **Bias-reducing intervention:** STEREOTYPE REPLACEMENT, COUNTER-STEREOTYPIC IMAGING, INDIVIDUATION, PERSPECTIVE-TAKING, or POSITIVE CONTACT.

This design lets us test whether training on intervention-labeled examples transfers beyond the exact surface form used during generation.

3.4 DEBIASITYOURSELF Instantiations

The primary implementation of DEBIASITYOURSELF trains or conditions a model on the intervention-labeled supervision above. Different instantiations answer different technical questions about where the supervision should enter the modeling pipeline.

Fine-tuned DEBIASITYOURSELF. The main trained variant fine-tunes LLAMA models with LoRA on intervention-labeled source/target triples.

We evaluate both intervention-specific adapters and all-intervention mixtures. The mixture is the central DEBIASITYOURSELF model because it learns a shared representation of bias-reducing behavior across multiple intervention types.

Inference-time DEBIASITYOURSELF. The inference-time variant conditions the model with the same intervention schema at evaluation time. This tests how much of DEBIASITYOURSELF can be activated without parameter updates and whether lightweight conditioning is sufficient for transfer.

Soft-prompt DEBIASITYOURSELF. Soft-prompt variants learn continuous prompt parameters for the intervention-labeled supervision. This isolates whether the debiasing behavior can be represented in a small learned conditioning vector.

Self-revising DEBIASITYOURSELF. Self-revising variants ask the model to first answer, then revise the answer under a selected intervention. This is an inference-time realization of the same structured supervision and is useful for tasks where a second pass can explicitly diagnose an unsupported group heuristic.

Ablation variants. In-context demonstrations, instruction-only prompts, chain-of-thought variants, sample-size variants, and source-form variants are ablations of DEBIASITYOURSELF. They

test whether performance depends on demonstrations, rationales, intervention labels, data volume, or the textual form of generated supervision.

4 Experimental Setup

4.1 Models

The main experiments use LLAMA-family causal language models, with LLAMA 8B as the primary model because it has the most complete bias, baseline, intervention, shot, and reasoning result matrix. We include QWEN and LLAMA-70B companion runs to test whether the same intervention interfaces behave similarly under alternative model families and scales. These companion results are used as robustness evidence rather than as the sole basis for aggregate claims. The camera-ready version should report the exact checkpoints, quantization settings, adapter merge procedure, and decoding settings for each model family.

4.2 Bias Evaluation

We evaluate on five primary bias benchmarks:

- **CrowS-Pairs**: stereotype preference should move toward 50.
- **StereoSet**: ICAT score should increase while stereotype score moves toward the neutral point.
- **BBQ**: accuracy should remain high while ambiguous and disambiguated bias scores move toward zero.
- **WinoBias**: gender-linked coreference bias should decrease.
- **WinoGender**: occupation-linked gender bias should decrease.

All main figures convert these benchmark-specific metrics into *bias error*, so lower values are consistently better: CrowS-Pairs stereotype preference and StereoSet SS overall are measured as distance from 50, while BBQ ambiguous/disambiguated bias scores, WinoBias absolute pro–anti gap, and WinoGender male–female pair disagreement are measured as distance from 0. The appendix includes companion-model plots and missing-data audits for QWEN and LLAMA-70B.

4.3 Baselines

We compare against a broad set of baselines implemented in the repository: DPO-style preference training, BIASEDIT, FairSteer, LFTF, PEFT, BiasFreeBench, BBA, CAL, Debias-NLG, MBIAS,

Debias-LLMs, Reduce Social Bias, DeCAP, and self-debiasing reprompting. These baselines cover training-time, representation-level, prompting, and inference-time mitigation families. The comparison is designed to test whether DEBIASITYOURSELF’s structured debiasing supervision provides better cross-benchmark behavior than methods defined primarily by a single mitigation mechanism. The final version should include a one-to-one mapping from every baseline label in the figures to its source paper, implementation, and hyperparameter setting.

4.4 Reasoning Evaluation

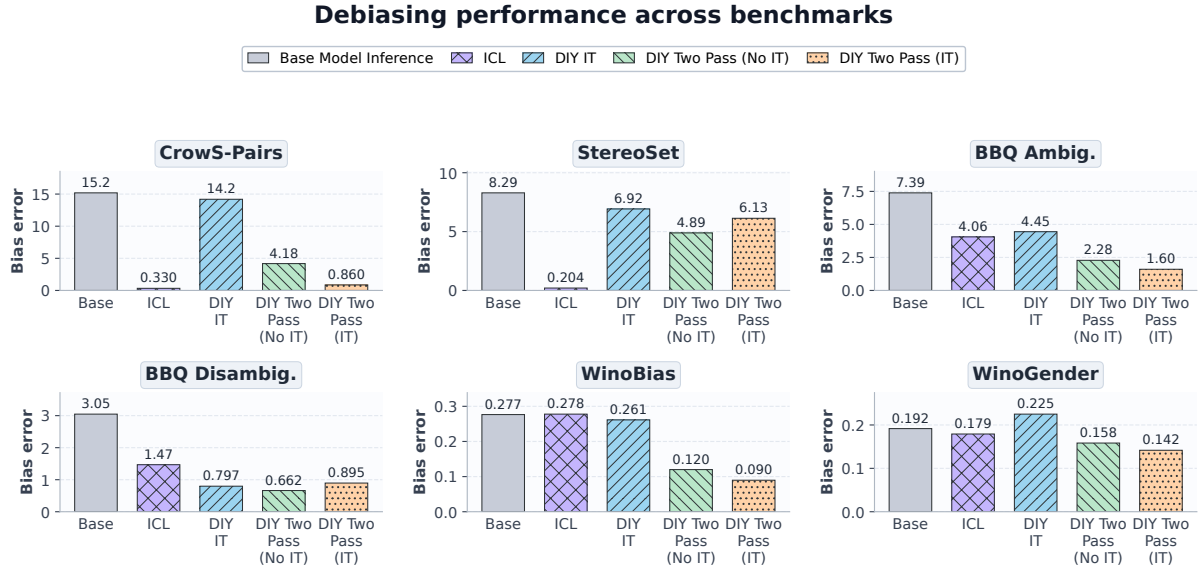
Bias reduction is not sufficient if the method degrades ordinary task competence. We therefore evaluate reasoning and knowledge tasks using Balanced COPA, ARC-Challenge, and ARC-Easy. These evaluations compare **BASE**, **ICL**, **DEBIASITYOURSELF IT**, **DEBIASITYOURSELF Two Pass (No IT)**, and **DEBIASITYOURSELF Two Pass (IT)**. For the primary LLAMA 8B setting, we use the full reasoning matrix in the main paper. For QWEN and LLAMA-70B, we report companion reasoning and bias–utility plots in the appendix.

5 Results

5.1 Debiasing performance

We first ask whether the main DEBIASITYOURSELF variants reduce bias across the full benchmark suite, rather than only under one favorable metric. Since the benchmarks report incompatible raw quantities, we plot each benchmark as a *bias error*: distance from the unbiased target. For CrowS-Pairs and StereoSet, the unbiased target is 50; for BBQ, WinoBias, and WinoGender, the unbiased target is 0. Lower values are therefore better in every panel. Unless otherwise stated, the main results in this section are from the LLAMA 8B run matrix, which is the only matrix currently complete across all three axes needed for the central comparison: bias benchmarks, baseline families, and reasoning benchmarks.

Figure 1 shows that DEBIASITYOURSELF improves the average normalized bias error over **BASE**, with the strongest aggregate reductions coming from inference-time intervention use. Across the six plotted bias metrics, **BASE** has a mean bias error of 5.73. **ICL** reduces this to 1.09, **DEBIASITYOURSELF IT** to 4.47,



Metrics: CrowS stereotype preference, StereoSet SS overall, BBQ ambiguous/disambiguated bias score, WinoBias pro-anti gap, and WinoGender male-female disagreement. Normalization: bars show distance from the unbiased target, 50 for CrowS/StereoSet and 0 for BBQ/Wino metrics; lower is better. ICL and DIY settings use all bias-reducing interventions.

Figure 1: Debiasing performance of **BASE**, **ICL**, and **DEBIASITYOURSELF** variants across the five primary bias benchmarks. Bars report benchmark-native metrics converted to bias error: CrowS-Pairs stereotype preference and StereoSet SS overall are measured as distance from 50, while BBQ ambiguous/disambiguated bias scores, WinoBias pro-anti gap, and WinoGender male-female pair disagreement are measured as distance from 0. Lower is better.

DEBIASITYOURSELF Two Pass (No IT) to 2.05, and **DEBIASITYOURSELF Two Pass (IT)** to 1.62. The pattern is not uniform across datasets, which is important: different benchmarks reward different behaviors, and **DEBIASITYOURSELF**'s gains are clearest when the model can explicitly condition on an intervention-guided procedure.

Result 1: **DEBIASITYOURSELF** reduces cross-benchmark bias error relative to **BASE**, with the strongest overall performance from **ICL** and **DEBIASITYOURSELF Two Pass (IT)**.

5.2 Beating baselines

Next, we compare **DEBIASITYOURSELF** against existing mitigation baselines. This comparison is intentionally broad: the baselines include prompting, data-centric, fine-tuning, preference-optimization, and representation-oriented approaches. The purpose is to test whether **DEBIASITYOURSELF** is competitive as a debiasing method, not merely better than the unmitigated base model.

Figures 2 and 3 show that no prior baseline dominates all datasets. Some baselines are strong on individual metrics, but their behavior varies substantially across benchmark families. **DEBIASITYOURSELF** is competitive

across this heterogeneous evaluation surface, and **DEBIASITYOURSELF Two Pass (No IT)** and **DEBIASITYOURSELF Two Pass (IT)** are especially strong on both mean bias-error and average-rank comparisons. This supports our framing of **DEBIASITYOURSELF** as structured debiasing supervision rather than as a benchmark-specific patch: the method remains competitive across pairwise likelihood, question answering, and coreference-style evaluations.

Result 2: **DEBIASITYOURSELF** is competitive with a broad set of existing debiasing baselines and is strongest when structured supervision is paired with intervention-guided inference.

5.3 Generalizability

We use interface robustness as one form of generalization: if **DEBIASITYOURSELF** is a reusable debiasing behavior, it should remain useful when the model receives the intervention signal through different inference-time contexts. We therefore compare the main instruction-tuned model with two-pass variants using 0, 1, and 2 demonstrations.

Figure 4 shows that **DEBIASITYOURSELF** is not tied to a single delivery mechanism. **DEBIASITYOURSELF IT** provides a learned interface to intervention-labeled supervision, while **DEBIASITYOURSELF Two Pass (No IT)** and

Model	Method	Mean bias error ↓	Mean reasoning accuracy ↑
LLAMA 8B	BASE	5.73	71.95
LLAMA 8B	ICL	1.09	70.21
LLAMA 8B	DEBIASITYOURSELF IT	4.47	70.81
LLAMA 8B	DEBIASITYOURSELF Two Pass (No IT)	2.05	74.83
LLAMA 8B	DEBIASITYOURSELF Two Pass (IT)	1.62	71.24
QWEN	BASE	4.64	87.27
QWEN	ICL	5.24	90.41
QWEN	DEBIASITYOURSELF IT	4.57	92.39
QWEN	DEBIASITYOURSELF Two Pass (No IT)	1.17	87.68
QWEN	DEBIASITYOURSELF Two Pass (IT)	1.77	92.96
LLAMA 70B	BASE	5.63	90.72
LLAMA 70B	ICL	2.71	89.34
LLAMA 70B	DEBIASITYOURSELF IT	5.85	89.98
LLAMA 70B	DEBIASITYOURSELF Two Pass (No IT)	2.08	89.84
LLAMA 70B	DEBIASITYOURSELF Two Pass (IT)	2.74	91.16

Table 2: Current aggregate summary across model families. Mean bias error averages the six normalized bias panels used in Figure 1; mean reasoning accuracy averages ARC-Challenge, ARC-Easy, and Balanced COPA. This table is intended as a compact navigation aid for the current draft rather than a replacement for per-dataset figures.

DEBIASITYOURSELF Two Pass (IT) activate the same intervention inventory at inference time. The shot-level comparison also reveals that additional demonstrations are not uniformly beneficial: some datasets improve with more demonstrations, while others are best under zero-shot inference. This suggests that the intervention signal itself is useful, but benchmark format and prompt sensitivity still matter.

Result 3: DEBIASITYOURSELF generalizes across training-time and inference-time interfaces, but the best shot setting depends on the benchmark and metric.

5.4 Bias vs. utility/reasoning

A debiasing method is not useful if it reduces bias by becoming evasive, invalid, or generally less competent. We therefore evaluate the same main variants on ARC-Challenge, ARC-Easy, and Balanced COPA, and then compare methods by their bias–utility tradeoff.

Figure 5 shows that DEBIASITYOURSELF Two Pass (No IT) and DEBIASITYOURSELF Two Pass (IT) preserve or improve reasoning accuracy in several settings, especially for ARC-Easy and Balanced COPA. DEBIASITYOURSELF IT alone is more mixed on ARC, which suggests that structured debiasing supervision is not automatically utility-neutral. ICL is also useful as a control: it can reduce bias strongly, but its reasoning behavior depends on benchmark format. Figure 6 summarizes the

overall tradeoff: methods vary not only in bias reduction but also in reasoning preservation. The desired region is low bias error with high reasoning accuracy; DEBIASITYOURSELF variants move toward this region more consistently than methods that reduce bias by degrading task performance.

Result 4: DEBIASITYOURSELF can reduce bias while preserving reasoning, but the tradeoff depends on the interface; DEBIASITYOURSELF Two Pass (No IT) and DEBIASITYOURSELF Two Pass (IT) give the strongest bias–utility profile in the current results.

5.5 Ablations

Finally, we test whether the five bias-reducing interventions matter. If DEBIASITYOURSELF were only generic debiasing data, then intervention-specific variants should behave similarly. Instead, we expect different interventions to specialize across task formats, with the all-intervention framing providing a robust aggregate method.

Figure 7 shows that intervention choice changes behavior. No single intervention is uniformly best across all datasets and interfaces. PERSPECTIVE-TAKING and POSITIVE CONTACT are strong in several two-pass settings, while INDIVIDUATION is especially useful when the task rewards attention to instance-specific evidence. DEBIASITYOURSELF IT has smaller but consistent gains across interventions. These patterns support the central design choice of DEBIASITYOURSELF: debiasing supervision should expose mul-

DIY compared with existing debiasing baselines



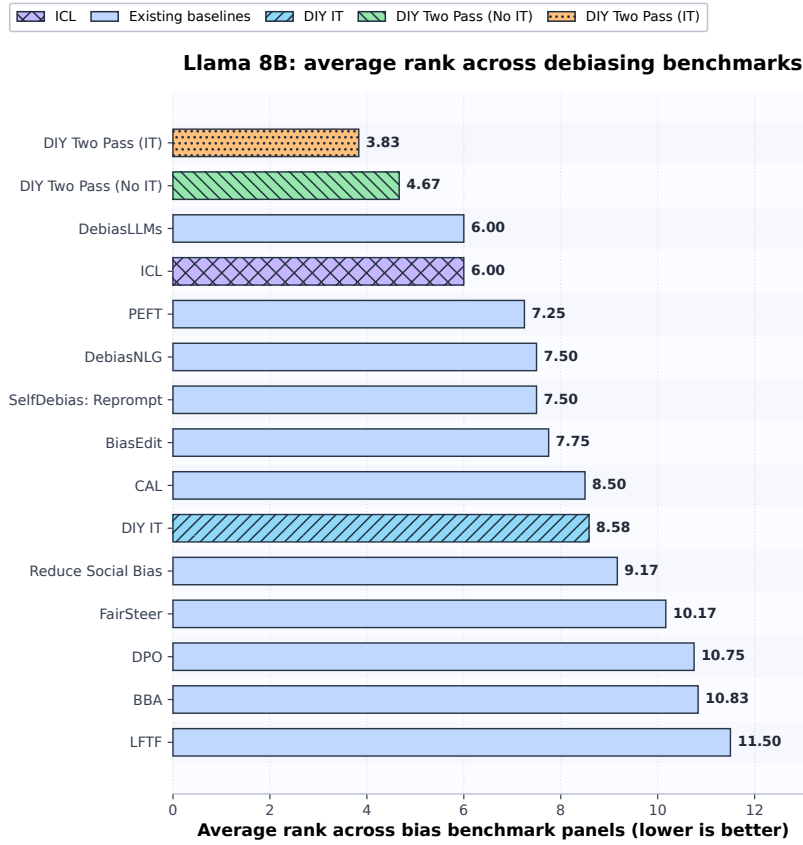
Metrics and normalization match the debiasing-performance figure; lower bias error is better. Baseline rows prefer Llama-8B results when available; DIY uses all bias-reducing interventions.

Figure 2: Comparison between DEBIASITYOURSELF and existing debiasing baselines. Each lollipop reports the same normalized bias-error metric used in Figure 1; lower is better. Circles denote **BASE** and prior baselines, while diamonds denote DEBIASITYOURSELF variants.

multiple reusable bias-reducing behaviors rather than force all examples through one generic instruction.

We include companion QWEN and LLAMA-70B plots in Appendix B. These plots do not replace the LLAMA 8B main results because the auxiliary and fine-grained matrices are incomplete in different ways, but the main debiasing and reasoning summaries now cover the same five method families. They are useful sanity checks because they expose where the DEBIASITYOURSELF interface appears model-sensitive.

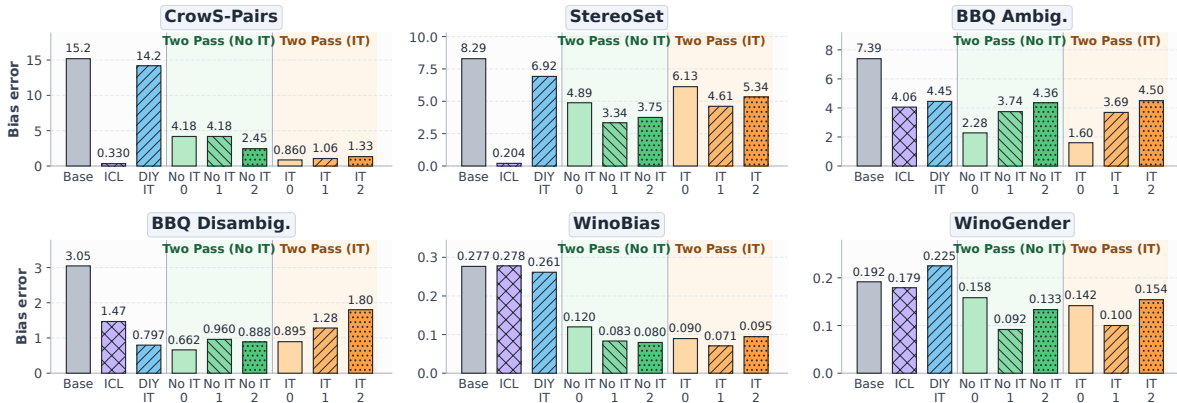
Result 5: The intervention inventory is not decorative; STEREOTYPE REPLACEMENT, COUNTER-STEREOTYPIC IMAGING, INDIVIDUATION, PERSPECTIVE-TAKING, and POSITIVE CONTACT specialize across benchmarks, and their mixture supports broader mitigation.



Ranks are computed independently within each bias panel using normalized bias error; rank 1 is best. Base model is excluded. Hatched rows use currently completed panels; pending rows have no rankable panel yet.

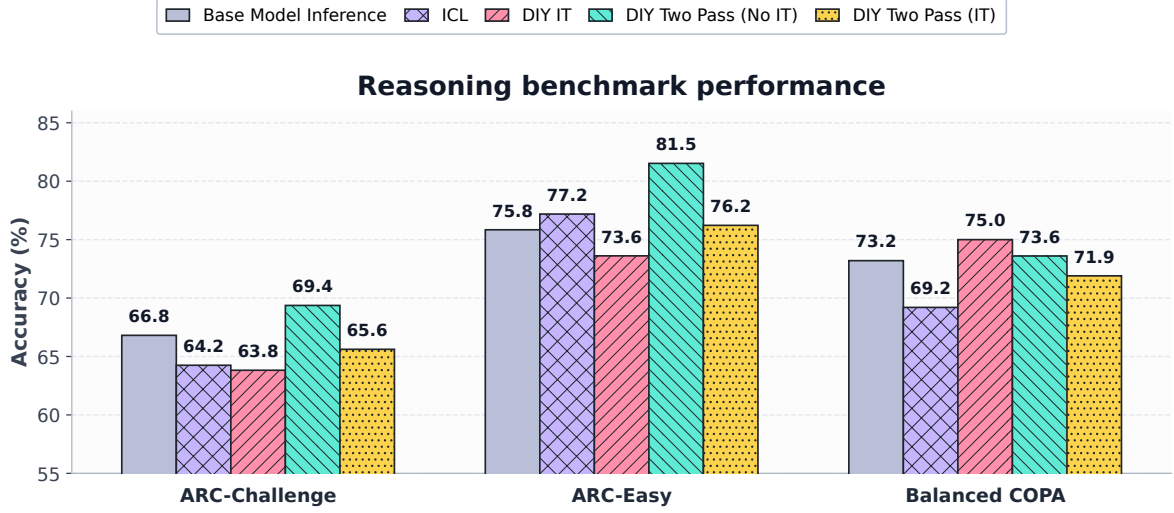
Figure 3: Average rank of DEBIASITYOURSELF and baseline methods across the available bias metrics. Lower rank is better. Ranking complements the lollipop comparison by reducing sensitivity to the scale of any one benchmark metric.

Debiasing performance by two-pass shot setting



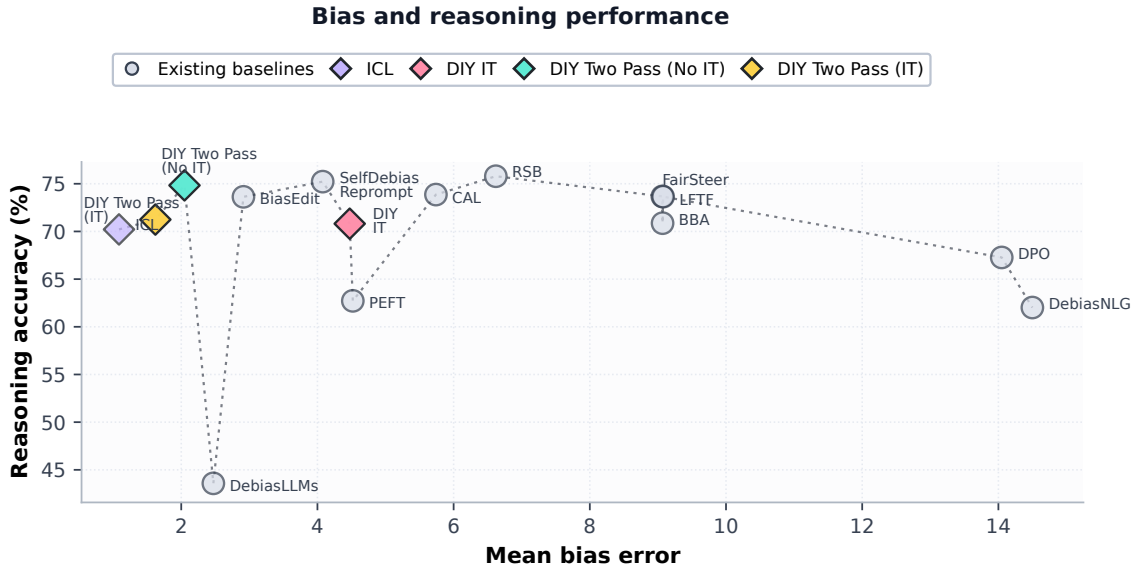
Metrics: CrowS stereotype preference, StereoSet SS overall, BBQ ambiguous/disambiguated bias score, WinoBias pro-anti gap, and WinoGender male-female disagreement. Normalization: bars show distance from the unbiased target, 50 for CrowS/StereoSet and 0 for BBQ/Wino metrics; lower is better. Shot labels 0/1/2 refer to demonstrations in the two-pass self-debiasing prompt.

Figure 4: Debiasing performance split by two-pass shot setting. **BASE** and **DEBIASITYOURSELF IT** are shown alongside **DEBIASITYOURSELF Two Pass (No IT)** and **DEBIASITYOURSELF Two Pass (IT)** with 0, 1, and 2 demonstrations. Metrics and normalization match Figure 1; lower is better.



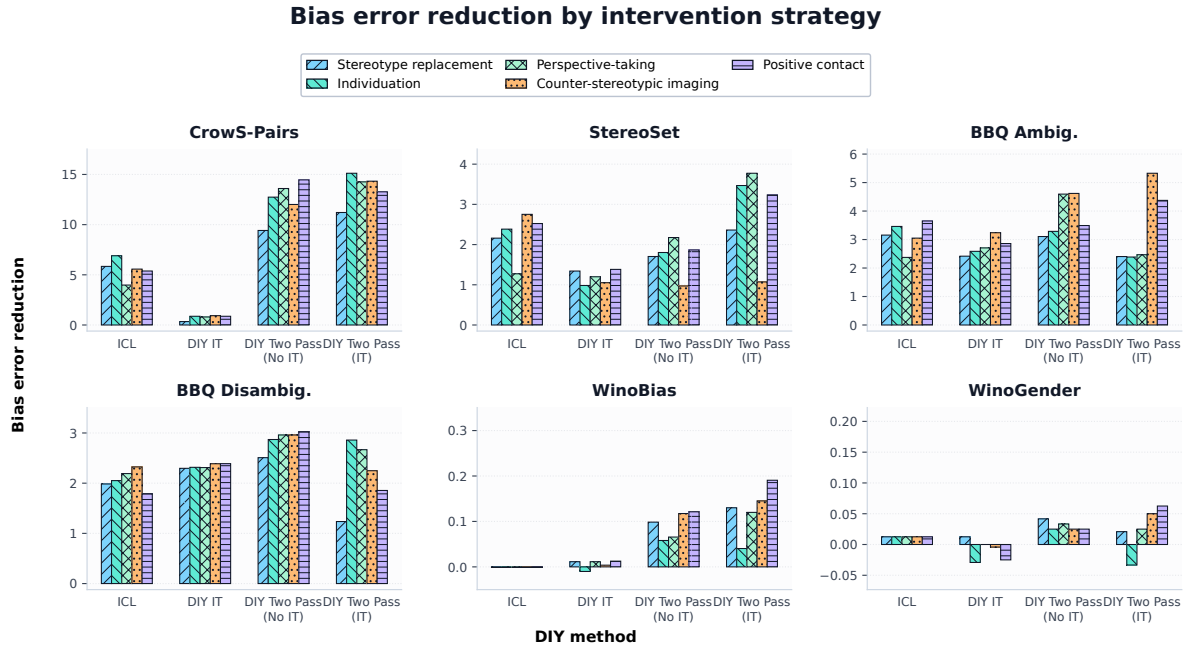
Metric: raw benchmark accuracy; higher is better. ICL and DIY settings use all bias-reducing interventions.

Figure 5: Reasoning benchmark performance for **BASE** and DEBIASITYOURSELF variants. Scores are raw accuracy on ARC-Challenge, ARC-Easy, and Balanced COPA; higher is better.



Bias error averages six normalized bias metrics; reasoning averages ARC-Challenge, ARC-Easy, and Balanced COPA. Lower bias and higher reasoning accuracy are better. All baseline points and all three DIY variants are shown.

Figure 6: Bias–utility tradeoff across baselines and DEBIASITYOURSELF variants. The x -axis is mean normalized bias error over the six bias panels; the y -axis is mean reasoning accuracy over ARC-Challenge, ARC-Easy, and Balanced COPA. Lower-left to upper-left movement indicates better bias mitigation without sacrificing reasoning.



Each subplot uses the benchmark-native metric normalized as bias error. Bars show absolute reduction from the base model on that metric; higher is better.

Figure 7: Intervention ablation across DEBIASITYOURSELF variants. Bars report bias-error reduction relative to (BASE) for each benchmark panel; higher is better. Colors correspond to STEREOTYPE REPLACEMENT, INDIVIDUATION, PERSPECTIVE-TAKING, COUNTER-STEREOTYPIC IMAGING, and POSITIVE CONTACT.

6 Analysis

6.1 What Does “Generalizable” Mean Here?

We define generalization along four axes. **Identity generalization** means the method is not tuned only to gender or race but applies across age, disability, nationality, religion, sexuality, socioeconomic status, and other axes. **Task generalization** means transfer across likelihood ranking, question answering, generation, and coreference-style evaluation. **Domain generalization** means improvement across benchmarks collected under different assumptions and annotation procedures. **Interface generalization** means that the same intervention-labeled supervision remains useful when instantiated through training-time and inference-time mechanisms.

The current evidence should be presented as partial generalization, not complete generalization. DEBIASITYOURSELF tests a broader surface than many single-benchmark mitigation methods, but all evaluations are still benchmark-mediated and

English-centered. The final paper should report where transfer fails as prominently as where it succeeds.

6.2 Why These Interventions?

The value of the socio-psychological source is not anthropomorphism. We do not argue that models have prejudice habits. Rather, prejudice-reducing interventions give us a compact design space for debiasing supervision. STEREOTYPE REPLACEMENT teaches explicit rejection and substitution. INDIVIDUATION teaches evidence use. PERSPECTIVE-TAKING teaches impact-aware inference. COUNTER-STEREOTYPIC IMAGING and POSITIVE CONTACT teach distributional diversification. Together, they define a structured supervision signal that is more precise than an ad hoc instruction such as “do not be biased.”

6.3 Failure Modes

We expect four recurring failure modes. First, some conditioned outputs may produce generic safety language rather than solving the task. Second, likelihood-based benchmarks can be distorted by verbose rationale text. Third, generated debiasing data can over-sanitize examples and reduce task diversity. Fourth, self-revision can improve bias metrics by changing answer style rather than by improving underlying evidence use. These risks motivate our use of multiple datasets and reasoning checks.

7 Discussion

DEBIASITYOURSELF is best understood as a structured debiasing supervision method with a psychologically motivated intervention inventory. It asks whether models can learn reusable mappings from biased inputs to evidence-grounded outputs, and whether those mappings transfer across bias surfaces. This framing separates the contribution from any single implementation mechanism and creates a more demanding evaluation standard: if the method is generalizable, it should show evidence across identities, benchmarks, and task types.

The strongest paper version should emphasize three results. First, DEBIASITYOURSELF produces competitive or strong bias mitigation across several benchmarks relative to existing baselines in the complete LLAMA 8B matrix. Second, the ablations show why the method works: intervention mixtures and structured supervision are more ro-

bust than unstructured instructions or single narrow interventions. Third, reasoning preservation is interface- and model-dependent: two-pass inference gives a strong bias–utility profile in the main matrix, and the companion QWEN and LLAMA-70B results show that the same interfaces remain competitive across model families, although some auxiliary cells are still missing.

This gives the paper a balanced but assertive claim. The method is not a universal debiasing solution. It is a generalizable structured supervision method that improves over narrower mitigation baselines by teaching models evidence-grounded bias-reducing behaviors.

8 Conclusion

We introduced DEBIASITYOURSELF, a structured supervision method for generalizable language-model bias mitigation. DEBIASITYOURSELF translates prejudice-reducing interventions from social psychology into model-facing bias-reducing interventions, generates intervention-labeled supervision across identity dimensions and instance types, and induces models to replace unsupported social-group heuristics with evidence-grounded outputs. The experimental design evaluates whether this method outperforms narrower debiasing baselines across benchmark formats while preserving reasoning performance. Current results show strong gains in the complete LLAMA 8B matrix and consistent companion evidence from QWEN and LLAMA-70B under the available bias and reasoning metrics.

Limitations

The current paper has several known limitations. First, several citation placeholders need verification before submission. Second, the evaluation is mostly English and benchmark-based, so claims about real-world bias mitigation should remain cautious. Third, generated biased examples can contain harmful stereotypes; the final paper must describe filtering, storage, and release decisions carefully. Fourth, some auxiliary companion-model and fine-grained cells remain incomplete, even though the main debiasing and reasoning plots are now populated for all three model families. Fifth, reasoning preservation is mixed and should not be overstated.

Ethical Considerations

DEBIASITYOURSELF intentionally generates and manipulates biased text in order to train and evaluate mitigation methods. This creates risks: synthetic examples can reproduce harmful stereotypes, benchmark improvements can create false confidence, and identity descriptors can be overgeneralized. We mitigate these risks by using the data for controlled research, evaluating across multiple identity dimensions, and explicitly distinguishing model behavior from human implicit attitudes. The final release plan should avoid publishing harmful generated examples without filtering, documentation, and access controls where appropriate.

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A Draft Appendix Checklist

The final appendix should include:

- full intervention prompts for all five interventions;
- data generation templates and filtering rules;
- model and LoRA hyperparameters;
- all benchmark metric definitions;
- full per-dataset result tables;
- per-intervention ablations;
- reasoning benchmark details;
- safety and data release documentation.

B Companion Model Results

The main paper uses the LLAMA 8B matrix for the primary narrative because it has the richest set of baseline, ablation, shot, and CI overlays. We additionally generate companion plots for QWEN and LLAMA-70B to inspect whether the same interfaces behave similarly under alternative model families and scales. These figures should be read as diagnostic evidence: the main debiasing and reasoning plots are populated, while some auxiliary and fine-grained cells remain missing and are tracked in figures/qwen/csv/ and figures/llama70b/csv/.

C Bootstrap Confidence Intervals

We compute bootstrap confidence intervals from available per-example prediction files where the outputs can be aligned across methods. The current CI plots are included as audit artifacts rather than as final statistical claims, because the final paper should verify alignment counts and metric recomputation for every dataset before reporting intervals in the main text.

D Planned Experiments and Analyses

Your current experiments cover the interfaces to DEBIASITYOURSELF, but the paper also needs experiments that prove the method itself is doing something structured and generalizable.

I would organize the story like this:

D.1 Current Experiments

In-context Learning. Shows whether bias-reducing interventions can be activated without training.

Instruction Tuning. Shows whether structured debiasing supervision can be learned.

Instruction Tuning + In-context / Two Pass. Shows whether trained models can further benefit from intervention-guided self-revision.

Reasoning Ability. Shows whether debiasing preserves general task competence.

Generalizability. Shows transfer across identities, datasets, domains, and task formats.

D.2 Major Missing Experiments

Intervention Ablation. This is essential.
Question:

Do the five bias-reducing interventions actually matter, or is any debiasing data enough?

Compare:

- all five interventions
- stereotype replacement only
- individuation only
- perspective-taking only
- counter-stereotypic imaging only
- positive contact only
- no intervention label
- generic “be unbiased” supervision

This supports the core claim that DEBIASITYOURSELF is structured supervision, not just more debiasing data.

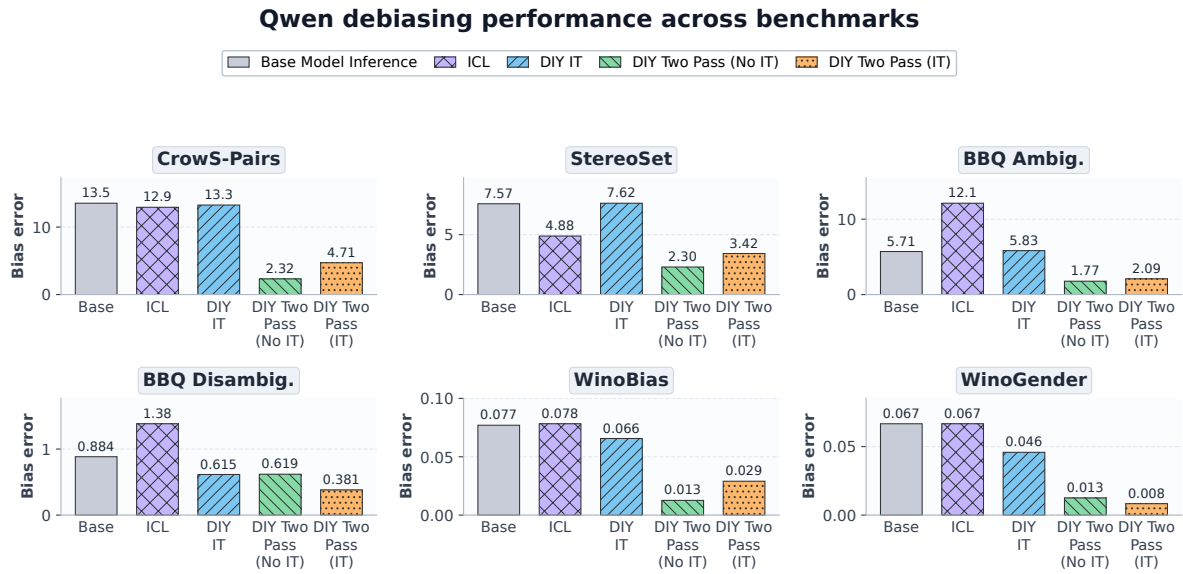
Intervention Label Ablation. Question:

Does labeling the intervention help?

Compare:

- input + debiased output
- input + intervention label + debiased output
- input + generic debias instruction + debiased output
- input + rationale + debiased output

This is very important. If labels help, the paper has a stronger technical story.



Metrics: CrowS stereotype preference, StereoSet SS overall, BBQ ambiguous/disambiguated bias score, WinoBias pro-anti gap, and WinoGender male-female disagreement. Normalization: bars show distance from the unbiased target, 50 for CrowS/StereoSet and 0 for BBQ/Wino metrics; lower is better. ICL and DIY settings use all bias-reducing interventions.

Figure 8: QWEN debiasing performance under the available bias metrics. The figure uses the same bias-error normalization as Figure 1 and includes **BASE**, **ICL**, **DEBIASITYOURSELF IT**, **DEBIASITYOURSELF Two Pass (No IT)**, and **DEBIASITYOURSELF Two Pass (IT)**.

Data Scale Ablation. Question:

How much supervision does DEBIASITYOURSELF need?

Compare:

- 100 examples
- 500 examples
- 1k
- 5k
- full data

This helps show DEBIASITYOURSELF is sample-efficient or that gains scale predictably.

Identity Holdout Generalization. This is probably the strongest generalizability experiment.

Train without one identity dimension, test on it.

Examples:

- train without religion, test on religion
- train without gender, test on gender
- train without race/ethnicity, test on race/ethnicity
- train without disability, test on disability

This proves DEBIASITYOURSELF is not memorizing identity terms.

Task Holdout Generalization. Train on some task formats, test on unseen task formats.

Examples:

- train on statements/dialogue/stories, test on QA
- train on generation-style examples, test on likelihood benchmarks
- train on QA, test on coreference
- train on sentence pairs, test on generation

This directly supports “generalizable across task types.”

Benchmark Holdout Generalization. Train or tune without examples resembling a benchmark, then test on that benchmark.

Examples:

- no coreference-like data, test WinoBias/WinoGender
- no QA-like data, test BBQ
- no pairwise sentence data, test CrowS-Pairs/StereoSet

This is stronger than just evaluating on five benchmarks.

Baseline Comparison by Mitigation Family.

Do not just compare methods in one table. Group baselines by type:

- prompting/self-debiasing

Qwen DIY compared with existing debiasing baselines



Metrics and normalization match the debiasing-performance figure; lower bias error is better. N/A marks unavailable Qwen metric files.

Figure 9: QWEN comparison with existing debiasing baselines. Metrics and normalization match the main LLAMA 8B baseline comparison.

- fine-tuning/data augmentation
- representation/steering
- preference optimization
- benchmark-specific debiasing

Then show DEBIASITYOURSELF is not only better than one weak baseline but competitive across mitigation families.

Utility Beyond Reasoning. Reasoning Ability is good, but reviewers may ask whether language quality or task ability drops.

Add:

- MMLU or ARC if already available
- HellaSwag / PIQA / BoolQ if easy
- perplexity or LM score if using likelihood models
- refusal / over-safety rate
- answer-format validity

Especially important because debiasing methods often “win” by becoming vague or refusing.

Over-Debiasing / Refusal Analysis. This is a key missing story.

Question:

Does DEBIASITYOURSELF reduce bias while still answering?

Measure:

- refusal rate
- generic safety response rate
- invalid answer rate
- “cannot determine” overuse on BBQ
- answer length inflation

This protects against the critique: “Your method just makes the model evasive.”

Qualitative Error Analysis. Include a small but sharp analysis.

Categories:

- unsupported group assumption removed
- individual evidence used correctly

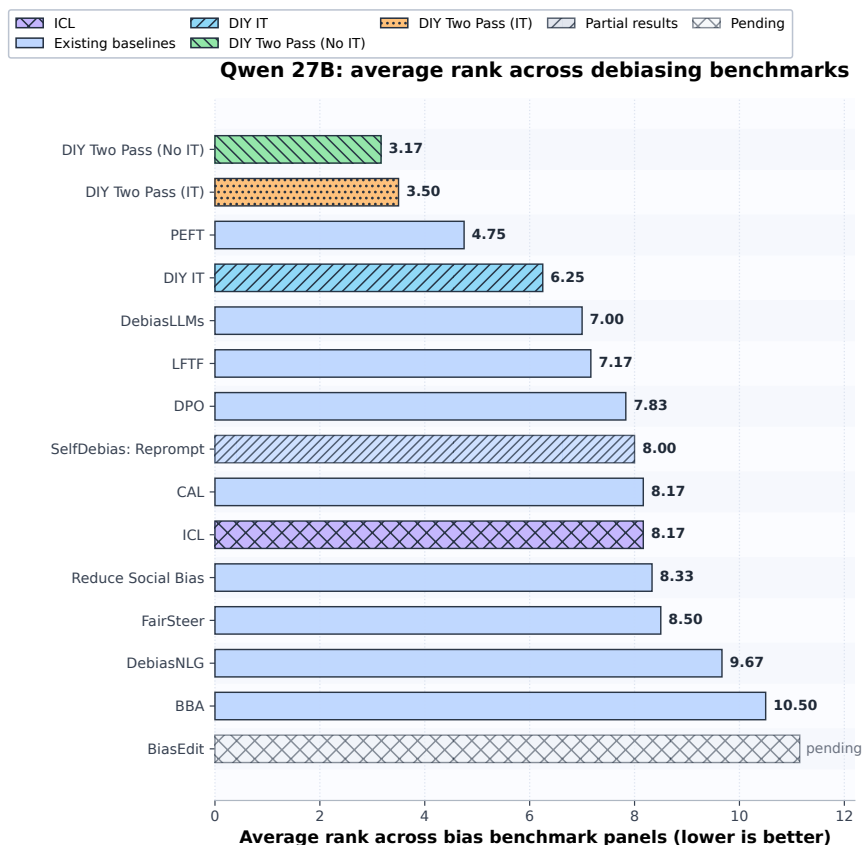


Figure 10: QWEN average-rank comparison across the available bias metrics. Lower rank is better.

- stereotype replaced but answer became generic
- perspective-taking caused verbosity
- model ignored task evidence
- benchmark artifact / format failure

This will help sell the intervention types as meaningful.

D.3 The Missing Story

Right now the story could still sound like:

We tried DEBIASITYOURSELF via ICL, tuning, two-pass, and checked generalization.

The stronger story is:

DEBIASITYOURSELF works because structured debiasing supervision teaches reusable bias-reducing behaviors, and those behaviors transfer across identities, tasks, and benchmarks better than generic debiasing or mechanism-specific baselines.

To support that, the must-have experiments are:

Intervention ablation. Do all five interventions matter?

Label/rationale ablation. Is structured supervision better than plain debiased examples?

Identity/task/benchmark holdout. Does it really generalize?

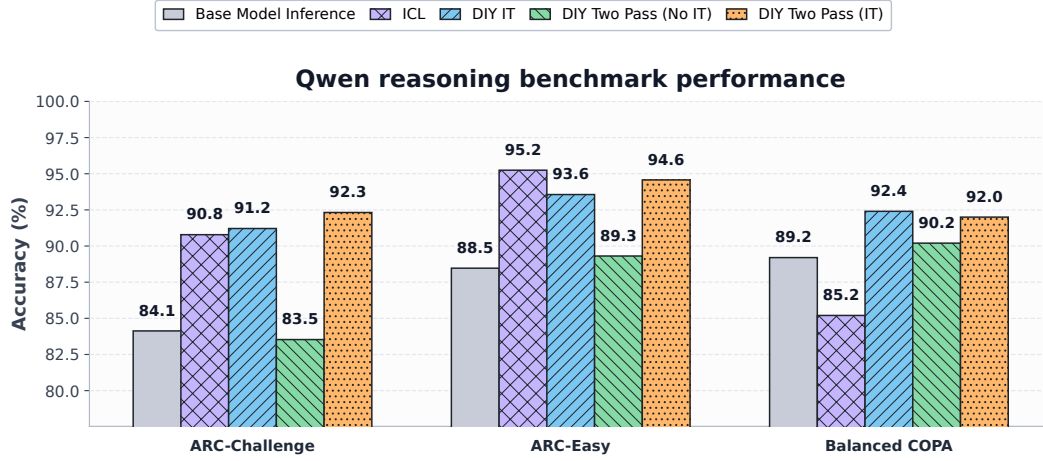
Over-debiasing analysis. Is the model still useful, or just evasive?

Baseline family comparison. Is DEBIASITYOURSELF better than strong alternatives, not just base model?

If space is limited, I would make the main paper experiments:

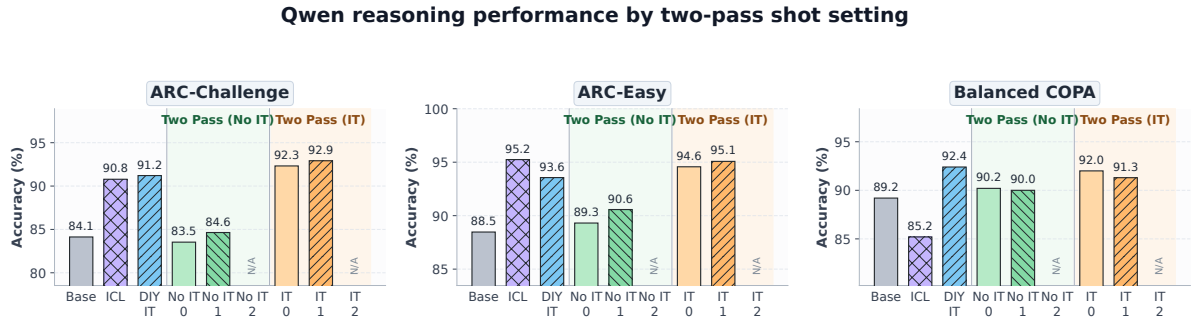
- Main benchmark comparison against baselines
- ICL vs instruction tuning vs two-pass
- Intervention and label ablations
- Generalization: identity holdout + task holdout
- Reasoning/utility + over-debiasing analysis

That would read like a strong EMNLP paper.



Metric: raw benchmark accuracy; higher is better. DIY IT uses the 500-example all-intervention Qwen checkpoint; two-pass settings use all interventions with zero demonstrations.

Figure 11: QWEN reasoning performance for (BASE) and DEBIASITYOURSELF variants. The reasoning matrix is complete for the main (BASE), (DEBIASITYOURSELF IT), (DEBIASITYOURSELF Two Pass (No IT)), and (DEBIASITYOURSELF Two Pass (IT)) variants.



Strategy setting: all interventions, IT checkpoint: all-version, Metric: raw benchmark accuracy; higher is better. Shot labels 0/1/2 refer to two-pass demonstrations; N/A marks unavailable Qwen reasoning files.

Figure 12: QWEN reasoning performance by two-pass shot setting. The added QWEN results primarily expand this reasoning analysis, showing how zero-, one-, and two-demonstration revision settings affect ARC and Balanced COPA accuracy.

D.4 Additional Experiments and Analyses

Additional experiments/analyses that would strengthen the paper:

1. Cross-Model Transfer. Train/design DEBIASITYOURSELF on one model, evaluate on others.

Examples:

- Llama 8B → Llama 70B
- Llama → Mistral/Qwen/Gemma
- closed model ICL only, if available

Question:

Are DEBIASITYOURSELF interventions model-specific, or do they transfer across architectures/scales?

This supports generality beyond one checkpoint.

2. Source Model Contamination / Generator

Ablation. If synthetic data is generated by an LLM, reviewers may ask whether gains come from the generator’s style.

Compare:

- data generated by GPT-4/Claude/etc.
- data generated by Llama
- human-written or benchmark-derived small set
- mixed generator data

Question:

Is DEBIASITYOURSELF robust to the data generator?

3. Synthetic Data Quality Ablation. Test filtering and quality controls.

Compare:



Bias error averages available normalized bias metrics; reasoning averages ARC-Challenge, ARC-Easy, and Balanced COPA. Lower bias and higher reasoning accuracy are better. Existing-baseline reasoning uses Qwen baseline utility runs when available.

Figure 13: QWEN bias–reasoning tradeoff. The plot mirrors Figure 6 and uses mean normalized bias error against mean reasoning accuracy.

- unfiltered generated data
- filtered data
- high-confidence only
- diverse identity-balanced data
- random subset with same size

This shows the pipeline matters, not just volume.

4. Intervention Selection / Routing. Right now each example has an intervention label. A stronger technical experiment:

Can the model choose which bias-reducing intervention to apply?

Setups:

- oracle intervention label
- predicted intervention label
- no intervention label
- all-intervention generic prompt

This would be very compelling if it works. It makes DEBIASITYOURSELF look like a structured model behavior, not a fixed prompt.

5. Multi-Intervention Composition. Some examples need more than one intervention.

Compare:

- single intervention

- two-intervention chains
- all-intervention mixture
- model-selected sequence

Question:

Do bias-reducing interventions compose?

This could be an appendix experiment unless results are strong.

6. Counterfactual Robustness. For each test example, swap identity terms and check prediction stability.

Metrics:

- prediction consistency across identity swaps
- sentiment/toxicity variance
- log-probability gap between identity variants
- answer flip rate

This is a clean, intuitive generalization test.

7. Intersectional Bias Evaluation. Evaluate combinations of identities, not only single-axis groups.

Examples:

- race + gender
- religion + nationality
- disability + gender

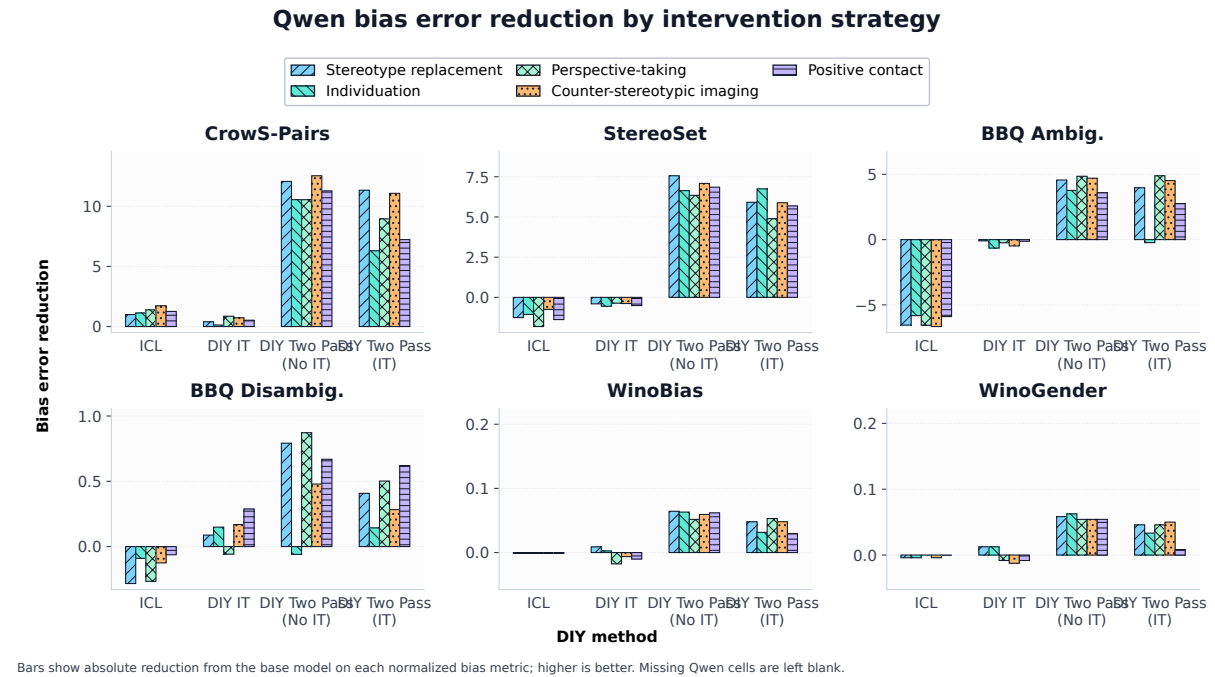
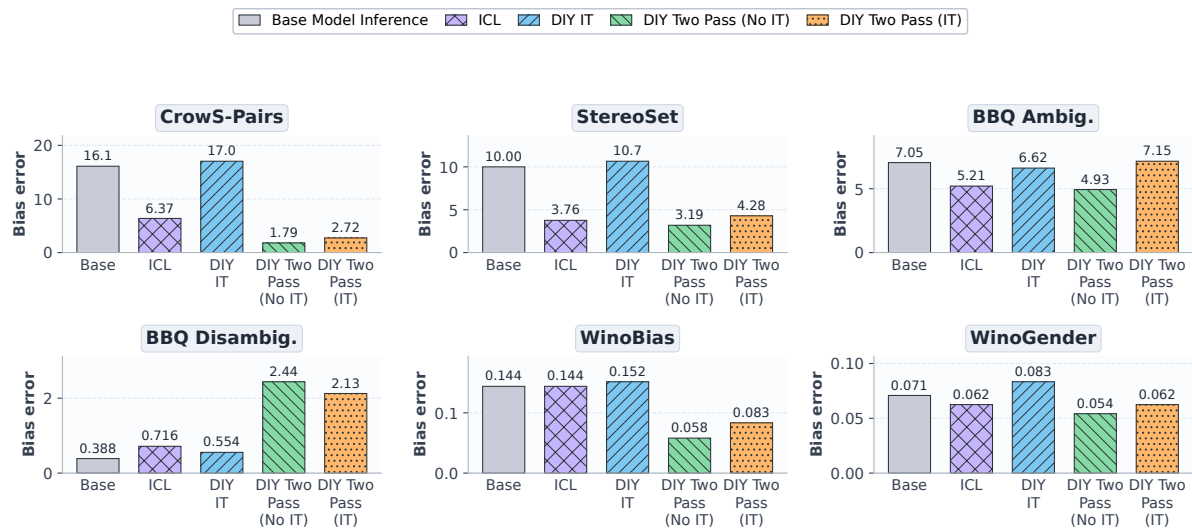


Figure 14: QWEN intervention ablation. Bars report bias-error reduction relative to the base model for the available QWEN benchmark panels.

1036	• age + socioeconomic status	Compare DEBIASITYOURSELF with baselines.	1059
1037	Question:	This is visually strong and methodologically clean. It shows whether DEBIASITYOURSELF gives a better tradeoff, not just lower bias.	1060
1038	Does DEBIASITYOURSELF generalize to intersectional identities not explicitly optimized for?		1061
1039			1062
1040			
1041	This would match your “across identities/dimensions” claim well.		1063
1042			1064
1043	8. Ambiguity Sensitivity. Especially for BBQ-style tasks.	10. Calibration Analysis. Bias mitigation can affect confidence.	1065
1044	A good debiased model should:	Measure:	1066
1045	• avoid stereotypes when context is ambiguous	• confidence on stereotypical vs anti-stereotypical examples	1067
1046	• use evidence when context is disambiguated	• calibration error before/after DEBIASITYOURSELF	1068
1047	Measure separately:	• entropy changes	1069
1048	• ambiguous bias score	• margin between answer choices	1070
1049	• disambiguated accuracy	Question:	1071
1050	• unknown-answer rate	Does DEBIASITYOURSELF reduce biased confidence, or does it just flip preferences?	1072
1051	• evidence-use accuracy		1073
1052	This prevents the model from always answering “unknown.”		1074
1053			1075
1054			
1055	9. Bias-Utility Pareto Frontier. Plot methods by:	11. Robustness to Prompt Wording. For ICL/inference-time DEBIASITYOURSELF:	1076
1056	• x-axis: utility/reasoning score	Compare multiple prompt phrasings:	1077
1057	• y-axis: bias reduction	• concise intervention label	1078
1058		• detailed intervention definition	1079
		• no social psychology wording	1080
		• generic debiasing instruction	1081
			1082

LLama-70B debiasing performance across benchmarks



Metrics: CrowS stereotype preference, StereoSet SS overall, BBQ ambiguous/disambiguated bias score, WinoBias pro-anti gap, and WinoGender male-female disagreement. Normalization: bars show distance from the unbiased target, 50 for CrowS/StereoSet and 0 for BBQ/Wino metrics; lower is better. ICL and DIY settings use all bias-reducing interventions.

Figure 15: LLAMA-70B debiasing performance under the available bias metrics. The plot mirrors Figure 1 and includes **BASE**, **ICL**, **DEBIASITYOURSELF IT**, **DEBIASITYOURSELF Two Pass (No IT)**, and **DEBIASITYOURSELF Two Pass (IT)**.

- intervention-specific prompt

Question:

Is DEBIASITYOURSELF robust, or prompt-fragile?

12. Robustness to Number/Order of Demonstrations. For ICL:

- 0-shot
- 1-shot
- 3-shot
- 5-shot
- shuffled demonstrations
- mismatched intervention demonstrations

This helps explain whether examples, labels, or both drive gains.

13. Negative Controls. Very important for reviewers.

Examples:

- random intervention labels
- shuffled debiased targets
- unrelated instruction labels
- generic moralizing labels
- same data without identity terms

If DEBIASITYOURSELF still works with random labels, the labels are not meaningful. If performance drops, your structure matters.

14. Adversarial / Stress-Test Prompts. Evaluate on harder bias-sensitive prompts:

- subtle stereotypes
- coded language
- negated stereotypes
- stereotype in quoted speech
- irrelevant identity mention
- identity genuinely relevant to answer

Question:

Does DEBIASITYOURSELF distinguish bias from legitimate identity context?

This is valuable because over-debiasing can erase relevant identity information.

15. Human Evaluation. If feasible, a small human eval would help.

Annotators rate:

- bias
- helpfulness
- evidence grounding
- over-refusal

Llama-70B DIY compared with existing debiasing baselines



Metrics and normalization match the debiasing-performance figure; lower bias error is better. N/A marks unavailable Llama-70B metric files.

Figure 16: LLAMA-70B comparison with existing debiasing baselines. The figure is useful as a scale companion to the main LLAMA 8B baseline comparison.

- fluency
 - whether identity was handled appropriately
- Even 200 examples can strengthen the paper.

16. Mechanistic / Representation Analysis. Optional, but technically appealing.

Examples:

- embedding shifts for identity terms
- attention to identity vs evidence tokens
- activation steering comparison
- logit attribution for stereotyped tokens
- whether DEBIASITYOURSELF reduces stereotype-token margins

This can make the paper look more technical, but only include if clean.

17. Data Diversity vs Intervention Diversity. Ablate:

- many identities, one intervention
- few identities, five interventions

- many task formats, one intervention
 - many interventions, one task format
- Question:

What drives generalization: identity diversity, task diversity, or intervention diversity?

This directly supports the story.

18. Seen vs Unseen Stereotype Templates. Train on some stereotype templates, test on unseen templates.

Example:

- train on occupation/gender stereotypes
- test on emotion/race stereotypes
- train on direct stereotypes
- test on subtle pragmatic stereotypes

Question:

Does DEBIASITYOURSELF memorize templates or learn debiasing behavior?

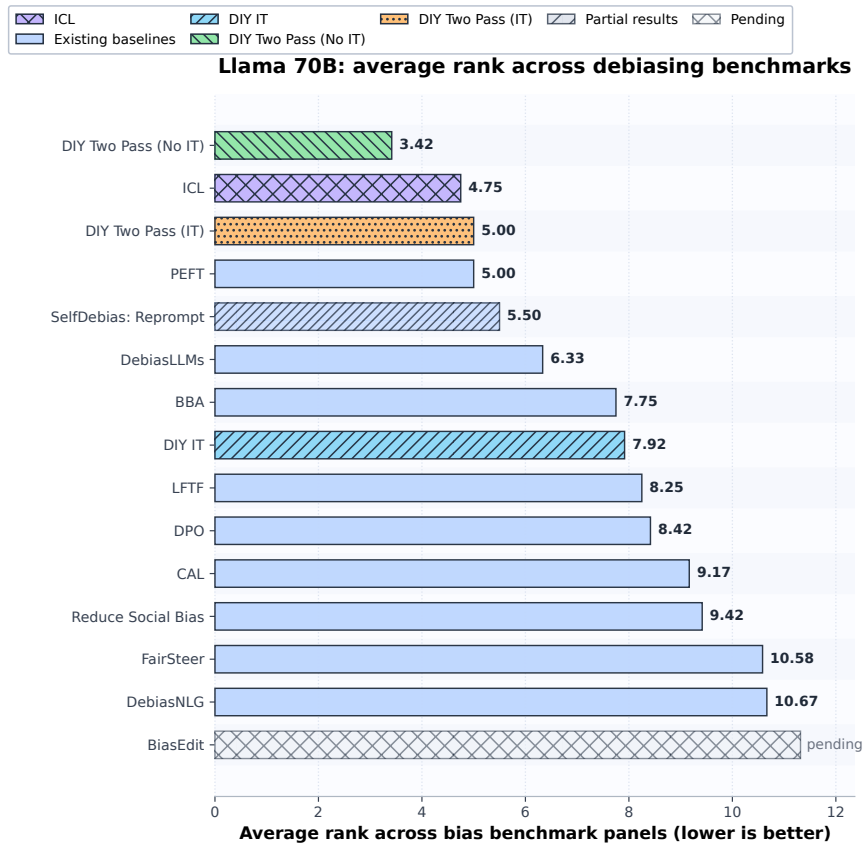


Figure 17: LLAMA-70B average-rank comparison across the available bias metrics. Lower rank is better.

19. Cross-Dataset Metric Agreement. Analyze whether improvements correlate across benchmarks.

Question:

Do current bias benchmarks measure the same thing?

This can be a useful analysis section:

- methods that improve CrowS but hurt BBQ
- methods that improve generation but not likelihood
- DEBIASITYOURSELF’s advantage is robustness across disagreeing metrics

20. Cost / Efficiency. Report:

- data size
- training time
- inference overhead
- two-pass overhead
- parameter count for LoRA/soft prompt

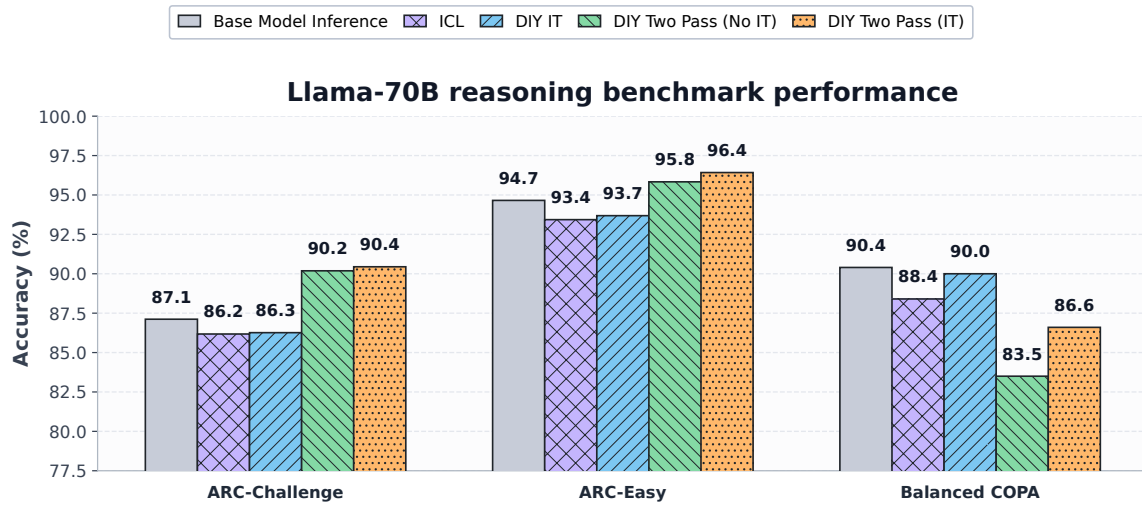
This matters because baselines may be cheaper or more expensive.

D.5 Most Valuable Additions

If you want the highest-impact additions, I would pick these:

- Negative controls: random labels, shuffled labels, generic debiasing.
- Identity/task/benchmark holdout: strongest generalization proof.
- Over-refusal and ambiguity sensitivity: prevents reviewer criticism.
- Bias-utility Pareto frontier: clean story figure.
- Intervention routing/selection: makes the method technically novel.
- Data diversity vs intervention diversity: proves what part of DEBIASITYOURSELF matters.

Those would make the paper much harder to dismiss as “psychology plus fine-tuning.”



Metric: raw benchmark accuracy; higher is better. Empty panels indicate that comparable Llama-70B reasoning metrics were not found in the current result folders.

Figure 18: LLAMA-70B reasoning performance for **BASE** and DEBIASITYOURSELF variants on ARC-Challenge, ARC-Easy, and Balanced COPA.



Bias error averages available normalized bias metrics; reasoning averages ARC-Challenge, ARC-Easy, and Balanced COPA. Lower bias and higher reasoning accuracy are better. Existing-baseline reasoning uses Llama-70B baseline utility runs when available.

Figure 19: LLAMA-70B bias–reasoning tradeoff. The plot mirrors Figure 6 and uses mean normalized bias error against mean reasoning accuracy.

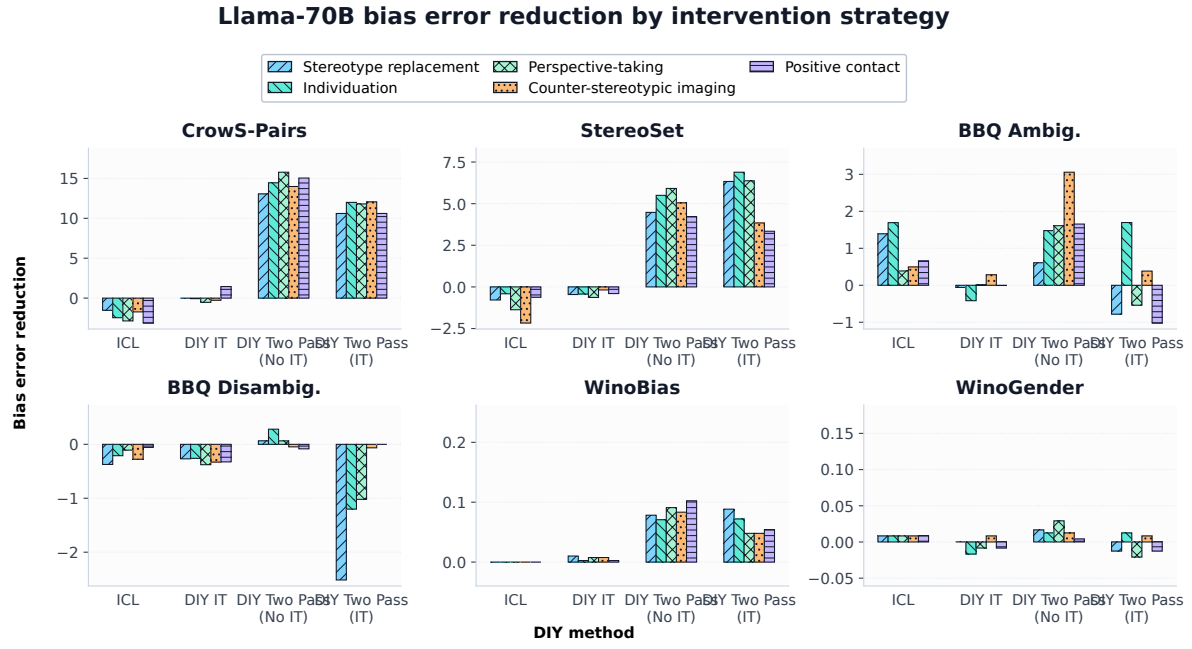


Figure 20: LLAMA-70B intervention ablation. Bars report bias-error reduction relative to the base model for the available LLAMA-70B benchmark panels.

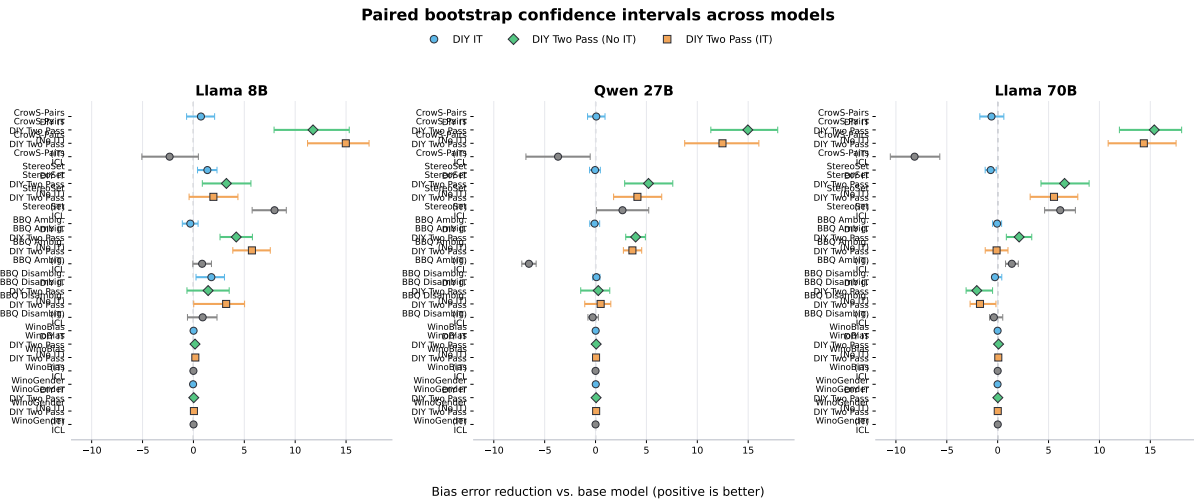


Figure 21: Bootstrap confidence intervals across model families for the currently aligned comparisons. Intervals are generated from existing output files and should be treated as draft uncertainty estimates until the final alignment audit is complete.